

作业4.

$$[(\text{Food} \Rightarrow \text{Party}) \vee (\text{Drinks} \Rightarrow \text{Party})] \Rightarrow [(\text{Food} \wedge \text{Drinks}) \Rightarrow \text{Party}]$$

(a) 枚举法.

Food, Party, Drinks, $(F \Rightarrow P) \vee (D \Rightarrow P)$, $(F \wedge D) \Rightarrow P$, 整个命题

T	T	T	T	T	T
F	T	T	T	T	T
T	F	T	F	F	T
T	T	F	T	T	T
F	F	T	T	T	T
F	T	F	T	T	T
T	F	F	T	T	T
F	F	F	T	T	T

故该语句是有效的、可满足的

(b) 左侧: $[(\text{Food} \Rightarrow \text{Party}) \vee (\text{Drinks} \Rightarrow \text{Party})]$

等价于 $(\neg \text{Food} \vee \text{Party}) \vee (\neg \text{Drinks} \vee \text{Party})$

等价于 $(\neg \text{Food}) \vee \text{Party} \vee (\neg \text{Drinks})$

右侧: $[(\text{Food} \wedge \text{Drinks}) \Rightarrow \text{Party}]$

等价于 $\neg (\text{Food} \wedge \text{Drinks}) \vee \text{Party}$

用德摩根定律, 则 $(\neg \text{Food}) \vee (\neg \text{Drinks}) \vee \text{Party}$.

则转换为合取范式CNF后: $[(\neg \text{Food}) \vee \text{Party} \vee (\neg \text{Drinks})] \Rightarrow [(\neg \text{Food}) \vee (\neg \text{Drinks}) \vee \text{Party}]$

则令 $(\neg \text{Food}) \vee \text{Party} \vee (\neg \text{Drinks}) = A$

$A \Rightarrow A$ 必然为真, 是有效的, 且是可满足的.

则语句是有效的, 是可满足的

(c) 归结法, 令左侧为A, 右侧为B, 则原命题 $A \Rightarrow B$

若要证明 $A \Rightarrow B$, 则需证明 $(A \wedge \neg B)$ 是不可满足的.

$A \wedge \neg B$ 即为 $[(\neg F) \vee P \vee (\neg D)] \wedge \neg [(\neg F \vee (\neg D) \vee P)]$
 $[(\neg F) \vee P \vee (\neg D)] \wedge [F \wedge D \wedge (\neg P)]$

则包含4个语句

① $(\neg F) \vee P \vee (\neg D)$

② F

③ D

④ $\neg P$

①、② 归结得: $P \vee (\neg D)$ ⑤

③、⑤ 归结得 P ⑥

④、⑥ 归结得空集, 故 $(A \wedge \neg B)$ 是不可满足的, 永假式

则原命题是有效的, 可满足的.